Woohyeon Byeon

woohyeon.byeon@kaist.ac.kr homepage whbyeon woohyeon-byeon

About & Research Interest

I am a Ph.D. student at KAIST, advised by Prof. Youngchul Sung. My main research interests are:

- **MARL & Mixed-Motive Games**: Multi-agent learning and systems, particularly in non-cooperative settings.
- **MORL** (Multi-Objective Reinforcement Learning): Sequential decision-making that balances trade-offs among competing objectives.

My broader goal is to understand how agents with diverse goals interact and make decisions, and to explore their behaviors in realistic environments.

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, S. Korea
Ph.D. Student in Electrical Engineering — Sep. 2024 – Present

- Advisor: Prof. Youngchul Sung

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, S. Korea
M.S. in Electrical Engineering — Sep. 2022 – Aug. 2024

- Advisor: Prof. Youngchul Sung

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, S. Korea
B.S. double major in Mathematical Sciences and Electrical Engineering (CUM LAUDE) — Mar. 2018 - Aug. 2022

Publications

C: Conference / **P**: Preprint / **W**: Workshop / **U**: Under-review ($*$: equal contribution, †: equal corresponding)

[C6] AGiR: Mitigating Gift Over-Reliance in Mixed-Motive Games — Sydney, Australia, Dec. 2026
Woohyeon Byeon, Seongmin Kim, Jiwon Jeon, Woojun Kim†, Yungchul Sung†

- NeurIPS 2026

[C5] Addressing Exogenous Variability in Cooperative Multi-Agent Reinforcement Learning — Sydney, Australia, Dec. 2026
Seongmin Kim, **Woohyeon Byeon**, Jiwon Jeon, Seungyul Han, Yungchul Sung

- NeurIPS 2026 (Spotlight)

[C4] MASTARS: Multi-Agent Sequential Trajectory Augmentation with Return-Conditioned Subgoals — Sydney, Australia, Dec. 2026
Jiwon Jeon, Myungsik Cho, Woojun Kim, Seongmin Kim, **Woohyeon Byeon**, Seungyul Han, Yungchul Sung

- NeurIPS 2026

[U1] Be My Tutor: On-Policy Co-Distillation for Mutual LLM Improvement via Peer Feedback
Woohyeon Byeon, Jiwon Jeon, Jeonghye Kim, Yungchul Sung

- Under-review

[C3] Constrained Multi-Objective Reinforcement Learning with Max-Min Criterion — Seoul, South Korea, Jul. 2026
Giseung Park*, Hyunyoung Nam*, **Woohyeon Byeon**, Amir Leshem, Yungchul Sung

- ICML 2026

[C2] Multi-Objective Reinforcement Learning with Max-Min Criterion: A Game-Theoretic Approach — San Diego, USA, Dec. 2025

Woohyeon Byeon, Giseung Park, Jongseong Chae, Amir Leshem, Yungchul Sung

- NeurIPS 2025

[C1] The Max-Min Formulation of Multi-Objective Reinforcement Learning: From Theory to a Model-Free Algorithm — Vienna, Austria, Jul. 2024

Giseung Park, **Woohyeon Byeon**, Seongmin Kim, Amir Leshem, Yungchul Sung

- ICML 2024

Projects & Experiences

Research Resident — Dec. 2025 - Feb. 2026

NAVER Cloud (K-AI Frontiers Program) — South Korea

- Contributed to the *K-Foundation Model Project*, participating in the Sovereign AI competition to develop independent foundational language models.
- Designed and constructed evaluation datasets to systematically benchmark and assess model capabilities.

Korea-Israel International Research — Sep. 2022 - Sep. 2024

Supported by the National Research Foundation of Korea and the Israel Ministry of Innovation, Science and Technology

- Collaborated with Prof. Amir Leshem's group at Bar-Ilan University.
- Developed a fair traffic signal control algorithm via multi-objective RL, published at ICML 2024.
- Developed memory- and cost-efficient, stable game-theoretic methods, published at NeurIPS 2025.
- Developed a max-min multi-objective RL method under constraints, published at ICML 2026.

Research Student — Winter 2020 - Spring 2021

Department of Mathematical Sciences, KAIST (Mentor: Prof. Donghwan Kim)

- Studied convergence of (stochastic) gradient methods in minimax optimization.

Honors & Scholarships

Dean's List, KAIST — Fall 2020

National Science & Engineering Scholarship (STEM), Korea Student Aid Foundation (KOSAF) — 2018 - 2021

Teaching Experiences

Fall 2023	**EE214 Machine Learning Basics and Practices**	Teaching Assistant
Spring 2023	**EE305 Introduction to Electronics Design Lab**	Teaching Assistant
Fall 2019	**MAS101 Calculus I**	Student Tutor
Spring 2019	**MAS102 Calculus II**	Student Tutor

Skills

Python, Pytorch, verl, huggingface, vllm